Anglo-Chinese Junior College

JC2 Biology Preliminary Examination
Higher 2

A Methodist Institution
(Founded 1886)

BIOLOGY | **9744/ 01**
Paper 1 Multiple Choice
16 September 2022
1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your Name and Index number in the Answer Sheet provided.

There are **thirty** questions in this section. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this Question Paper.
The use of an approved scientific calculator is expected, where appropriate.

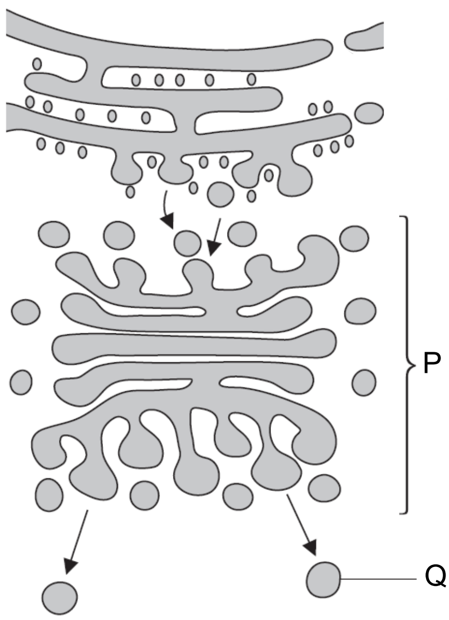
1 Different processes occur in different organelles in a plant cell.

1 transcription of circular DNA
2 movement of ions across protein channels
3 breaking of covalent bonds by hydrolysis
4 polymerisation of monomers containing nitrogen

Which processes occur in these three organelles?

	mitochondrion	chloroplast	nucleus
A	1, 2, 3, 4	1, 2, 3, 4	2, 3, 4
B	2, 4	2, 4	1, 3, 4
C	1, 3	4	2, 3
D	1	1	3, 4

2 The diagram shows some cell organelles involved in the formation and transport of trypsin.



What are the roles of organelles P and Q in the formation and transport of trypsin?

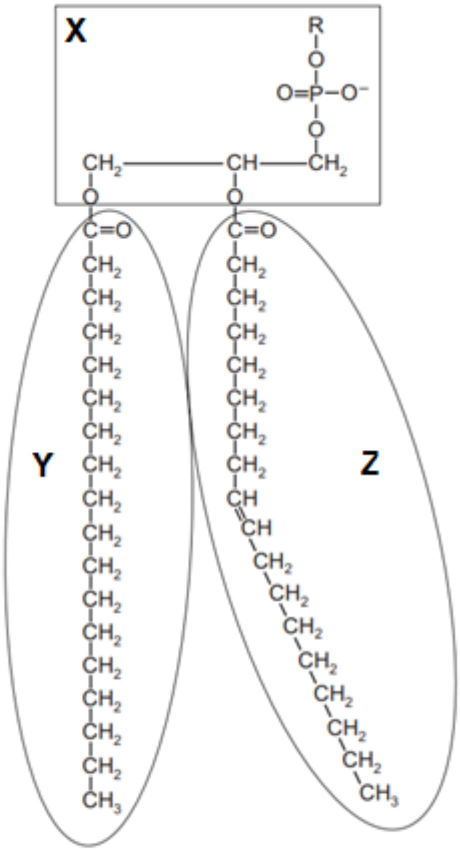
A P synthesises trypsin polypeptides on its surface and transports it from cisterna to cisterna before pinching off at the trans face as Q.

B Q has an acidic environment that prevents the activation of trypsin polypeptides modified in P, until it reaches the next organelle.

C Q transports trypsin polypeptides synthesised by free ribosomes to P for glycosylation.

D Trypsin polypeptides undergo post-translational chemical modification in the lumen of P, while Q, which contains the trypsin enzyme, fuses with the cell surface membrane.

3 The diagram shows a phospholipid molecule divided into three regions, X, Y and Z. In region X, R represents a range of possible chemical groups.



Which statements about regions X, Y and Z are correct?

1. Regions Y and Z are linked to region X via ester bonds.
2. Regions Y and Z contain the same number of carbon atoms, resulting in a very fluid cell surface membrane.
3. R, in region X, can be a sulfhydryl group which forms disulfide bonds with other sulfhydryl groups.
4. Region Z results in the formation of a kink in the phospholipid molecule.

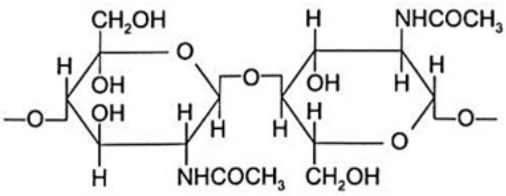
A 1, 2 and 3

B 1, 3 and 4

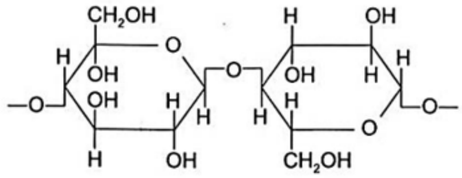
C 1 and 4 only

D 3 and 4 only

4 The diagrams show short sections of some common and modified polysaccharides.



1



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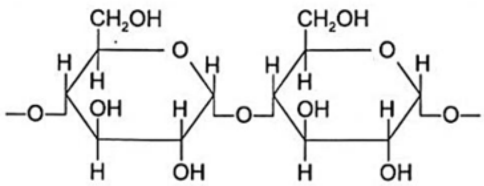

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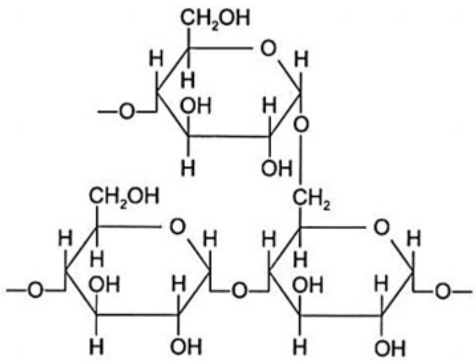


4

Which statement is correct?

A 1 is found in cellulose as β-1,4-glycosidic bond is present.

B 2 is found in amylopectin, which is a helical molecule, allowing for extensive coiling and entangling.

C 3 is found in cellulose, which is an unbranched and straight chain of β-glucose monomers.

D 4 is found in glycogen, which is made of chains of α-glucose linked by α–1,4–glycosidic bonds and α–1,6–glycosidic bonds at the branches.

5 Which comparative statements about collagen and haemoglobin are correct?

1 Collagen is more regular in structure.
2 Haemoglobin is less sensitive to changes in pH.
3 Haemoglobin is more soluble in water.
4 Collagen is more resistant to high temperatures.

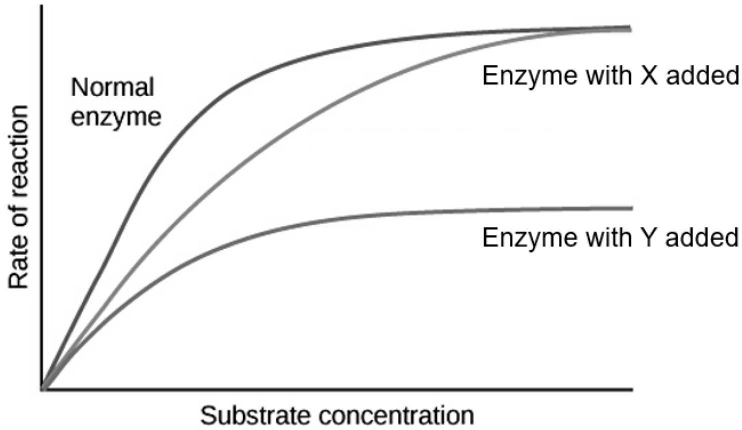
A 1, 2 and 3

B 1, 2 and 4

C 1, 3 and 4

D 2, 3 and 4

6 The effects of two different enzyme inhibitors, X and Y, were investigated.



Which statement best describes inhibitor X or Y?

A Inhibitor X could be a reversible, non-competitive inhibitor.

B In the presence of inhibitor X, V_{max} is the same as compared to without any inhibitor as X does not compete with the substrate for the active site of the enzyme.

C Inhibitor Y could be an irreversible, competitive inhibitor.

D In the presence of inhibitor Y, the rate of product formation eventually reaches zero at higher substrate concentration.

7 Which statement explains why induced pluripotent stem cells (iPSCs) are suitable for research and medical applications?

A They can be stimulated by chemical signals to express particular genes and give rise to specific cell types.

B They are specialised cells that can differentiate to give rise to almost any type of cell in the body.

C They can differentiate to a limited range of cells as they only have the genes required for a particular cell line.

D They lose genetic information as they divide, making it easy to reprogram them to differentiate into specific cell types.

8 How many statement(s) correctly describe the lambda bacteriophage?

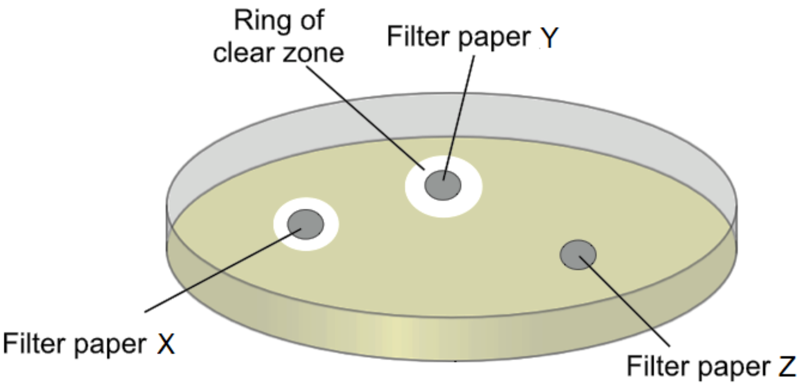
1 It has tail fibres that adsorb to specific proteins on bacteria.
2 The genome is made up of single-stranded DNA.
3 A phospholipid bilayer surrounds the nucleocapsid.
4 Its capsid contains a prophage.

A 0 **B** 1 **C** 2 **D** 3

[Turn over

9 An experiment was carried out to investigate the effect of bacteriophages and antibiotics on *E. coli*. Two strains of *E. coli* were used where one of the strains was resistant to antibiotics.

Three discs of filter papers were soaked with either temperate phages, virulent phages or antibiotics and placed on an agar plate spread with one strain of *E. coli*. The plate was then incubated at 37°C for 10 hours and rings of clear zones were observed. The diagram below shows the results.



The same experiment was repeated on a second agar plate containing the other strain of *E. coli* and there was only one ring of clear zone around filter paper Y.

Which explanation could be deduced from the experiment?

A Filter paper X was soaked with phages that did not initiate an immediate lysis of *E. coli* cells, resulting in a smaller clear zone around filter paper X than filter paper Y.

B Filter paper Y was soaked with phages that could integrate its genome into dividing *E. coli* cells, resulting in a larger clear zone around filter paper Y than filter paper X.

C Filter paper Z was soaked in antibiotics, resulting in the absence of a clear zone.

D Filter paper X was soaked in antibiotics but the strain of *E. coli* used in the second plate was resistant to antibiotics, resulting in no clear zone around filter paper X.

10 The following events occur during transcription.

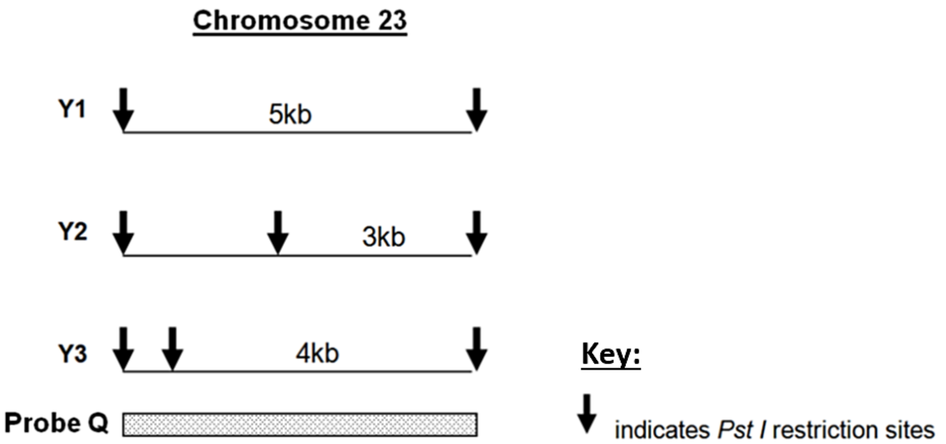
1 Bonds break between complementary bases.
2 Bonds form between complementary bases.
3 Sugar-phosphate bonds form.
4 Free nucleotides pair with complementary nucleotides.

Before the mRNA molecule leaves the nucleus, which events will have occurred twice?

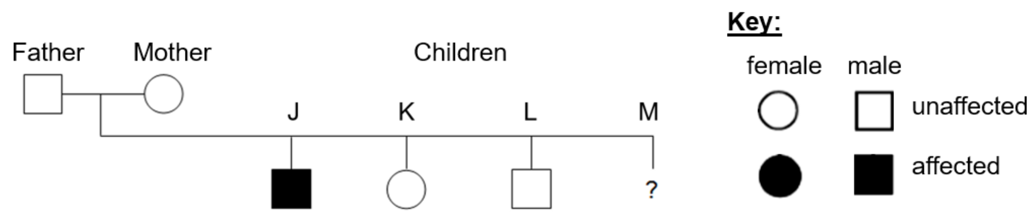
A 1, 2, 3 and 4 **B** 1, 2 and 3 only **C** 2, 3 and 4 only **D** 1 and 2 only

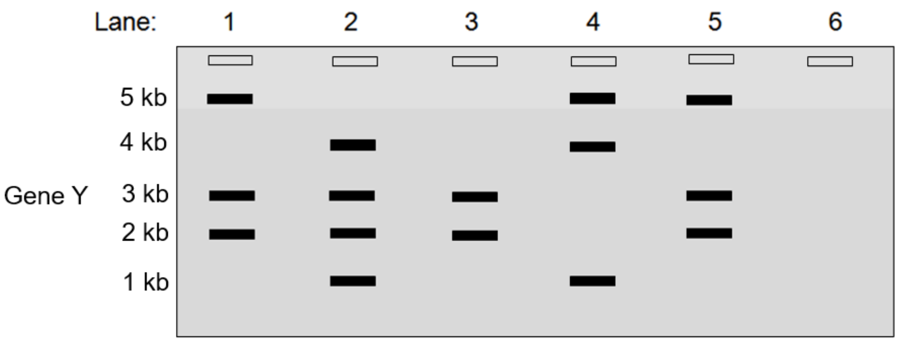
11 The DNA of a family affected by a rare autosomal disease was analysed using gel electrophoresis and Southern blotting. Gene Y, which is found on chromosome 23, is responsible for the disease.

Restriction enzyme *Pst* I was used to digest gene Y that has three different alleles, Y1 to Y3. Probe Q was used to identify the restriction fragments of the alleles of gene Y.



A phenotypically normal couple had three children, J, K, L, with only one of them affected by the disease. They recently gave birth to a fourth child, M.





Which conclusion could be made from the analysis?

A The father is a homozygote.

B The genotype of child J is Y2Y2.

C The bands present in lane 6 could be 2 kb, 3 kb and 4 kb.

D Child M will not be affected by the disease.

12 The flowchart shows how small interfering RNA (siRNA) affects the expression of a particular target gene.

A strand of siRNA combines with a protein to form an siRNA-protein complex.

↓

The siRNA-protein complex attaches to an mRNA molecule that codes for a particular protein.

↓

The siRNA-protein complex breaks the mRNA molecule down into smaller pieces.

Which statements are consistent with the information provided?

1 The siRNA-protein complex attaches to an mRNA molecule coding for a particular protein because the base sequence of siRNA is complementary to the specific mRNA molecule.

2 The presence of siRNA will increase the expression of the target gene because there will be an increase in the number of mRNA molecules after the siRNA-protein complex binds to it.

3 The expression of the target gene is affected by siRNA as the complete target protein is no longer synthesised after the mRNA is cut into pieces.

4 siRNA may be useful in treating genetic diseases due to mutations because the gene will not be transcribed.

A 1, 2 and 4 **B** 1 and 3 only **C** 2 and 4 only **D** 3 and 4

13 The ends of a eukaryotic chromosome contain a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in the shortening of telomeric DNA.

What is a consequence of the loss of repeating DNA sequences from the telomeres?

A The cell will begin the synthesis of different proteins.

B The cell will begin to differentiate as a result of the altered DNA.

C The number of mitotic divisions the cell can undergo will be limited.

D The production of mRNA will be reduced.

14 The life cycle of a fly includes a transition from the larval to the pupal stage. When the larva is fully grown, it changes into a pupa that does not feed. In the pupa, the tissues that made up the body of the larva are broken down. New adult tissues are formed from the substances obtained from these broken-down tissues and from substances that were stored in the body of the larva.

The table shows the mean concentration of RNA in fly pupae at different ages.

age of pupa as percentage of total time spent as a pupa	mean concentration of RNA / μg per pupa
0	20
20	15
40	12
60	17
80	33
100	20

What is a possible explanation for the change in RNA concentration in the pupa at different ages?

A From 0 to 40% of the time spent as a pupa, RNA concentration decreases because it is broken down together with the tissues that are broken down.

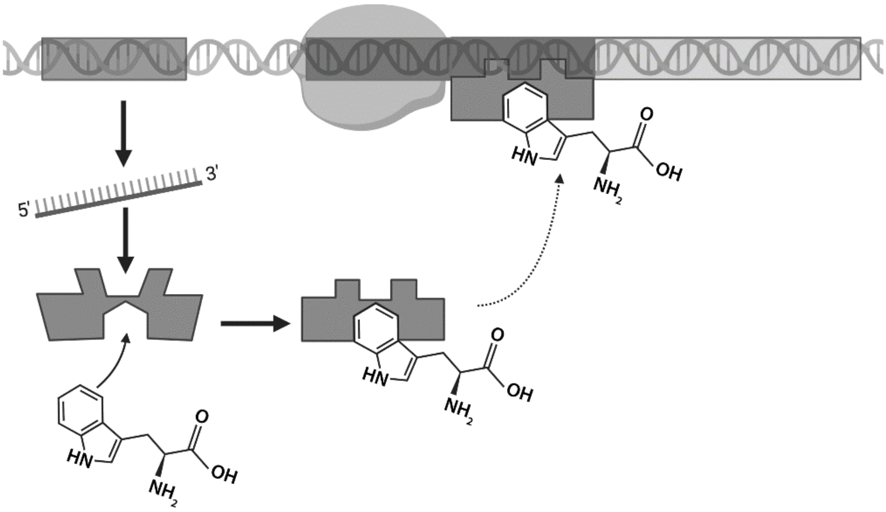
B From 0 to 100% of the time spent as a pupa, RNA concentration increases because genes are transcribed to produce more RNA for translation into proteins to form new adult tissues.

C From 60 to 80% of the time spent as a pupa, RNA concentration increases as a result of activators binding to promoters of genes that code for proteins required for the formation of new adult tissues.

D Overall concentration of RNA stayed the same throughout the time spent as a pupa because most RNAs are broken down when the pupa changes into an adult fly as no new protein needs to be synthesised.

[Turn over

15 The diagram shows the interaction of molecules with a segment of DNA which includes an operon.



molecule X

Which statement is **not** true?

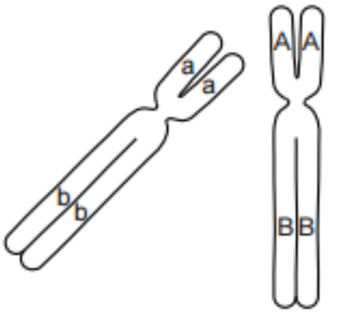
A The protein that molecule X binds to is originally inactive.

B The operon is negatively controlled by a repressor.

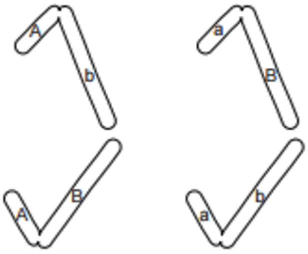
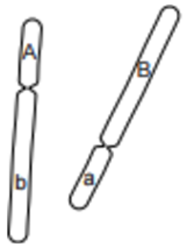
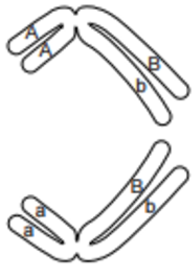
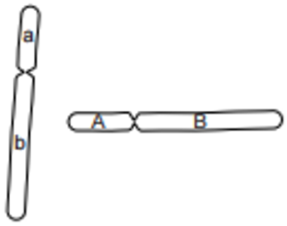
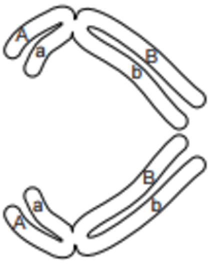
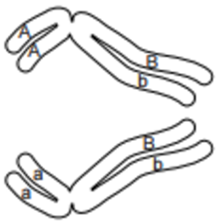
C This metabolic pathway is regulated by the process of feedback inhibition.

D The absence of molecule X allows the synthesis of proteins involved in catabolism.

16 The figure shows the location of two genes, A/a and B/b, on two homologous chromosomes in early prophase I of meiosis in an animal cell.

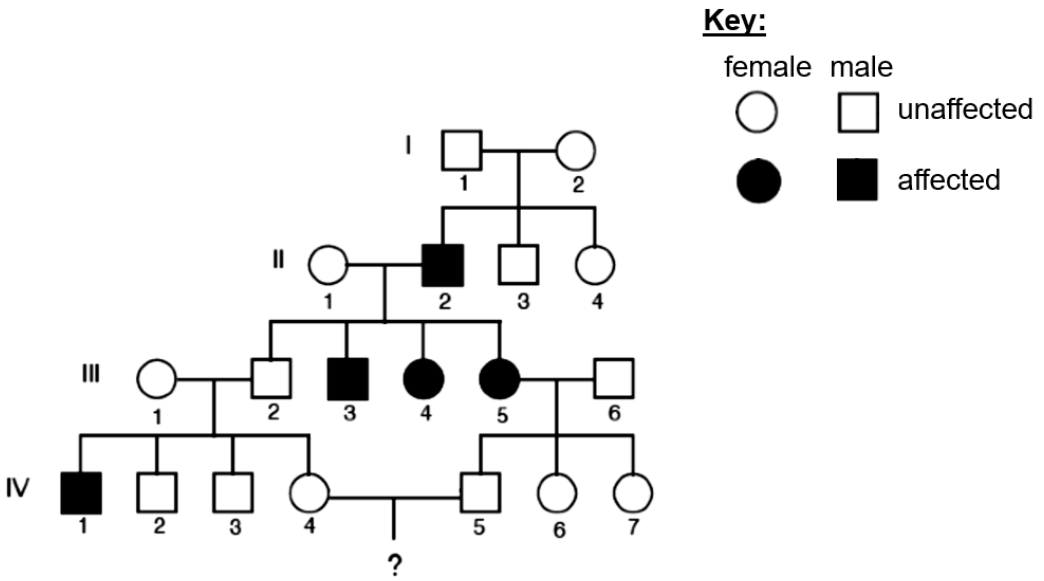


Which row is a possible representation of these chromosomes as they progress from anaphase I to prophase II?

	anaphase I	prophase II
A		
B		
C		
D		

[Turn over

17 The pedigree shows the inheritance of a trait over three generations in an extended family.



What is the probability that individuals IV–4 and IV–5 have a son who has the trait?

A 1 in 4 **B** 1 in 6 **C** 1 in 8 **D** 1 in 12

18 In the inheritance of feather colour in chickens, individuals carrying the dominant allele, W, have white plumage even if they also carry the dominant allele, C, for coloured plumage.

White Leghorn chickens have the genotype WWCC and white Wyandotte chickens have the genotype wwcc. A white Leghorn is crossed with a white Wyandotte and the F2 generation yielded 27 chickens with white plumage and 7 chickens with coloured plumage. The expected ratio for this cross was 13 : 3.

A chi-squared test was performed to test the significance of the difference between the observed and expected results. It was found that there was no significant difference, at a 95% confidence level, between the observed and expected results.

table of chi-squared values					
degrees of freedom	probability P (%)				
	95	80	50	20	5
1	0.00393	0.0642	0.455	1.642	3.841
2	0.103	0.446	1.386	3.219	5.991
3	0.352	1.005	2.366	4.642	7.816

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = observed result
E = expected result
v = n–1

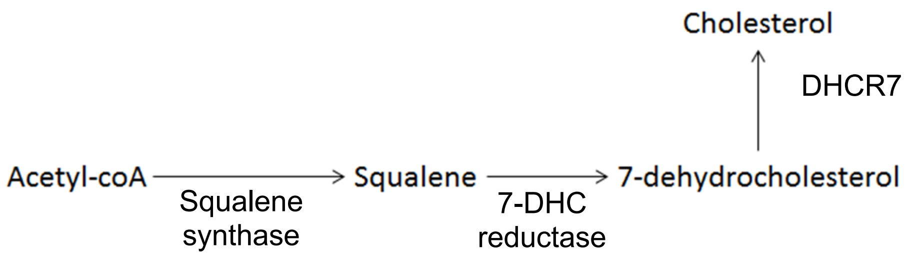
Using the equation and table above, which value would be close to the calculated chi-square value for this experiment?

A 0.098 **B** 3.34 **C** 3.88 **D** 7.52

19 A gene found on chromosome 11 codes for an enzyme called 7-dehydrocholesterol reductase (DHCR7). This enzyme is involved in the biosynthesis of cholesterol. Inheritance of two mutated alleles results in a disease known as Smith-Lemli-Opitz syndrome, which is characterised by a reduced level of cholesterol.

Another gene found on this chromosome codes for the protein alpha-tectorin (TECTα), which is a major component of the tectorial membrane in the inner ear. A dominant mutation results in a disease known as non-syndromic deafness.

Apart from DHCR7, other enzymes required in the biosynthesis of cholesterol are squalene synthase and 7-DHC reductase which are coded for by genes found on other chromosomes.



Which statement cannot be concluded?

A The *DHCR7* gene and *TECTα* gene do not assort independently.

B A person who is heterozygous for both *DHCR7* and *TECTα* genes suffers from non-syndromic deafness but not Smith-Lemli-Opitz syndrome.

C The production of cholesterol from acetyl-CoA is influenced by gene interactions.

D The amount of cholesterol that can be produced shows discontinuous variation.

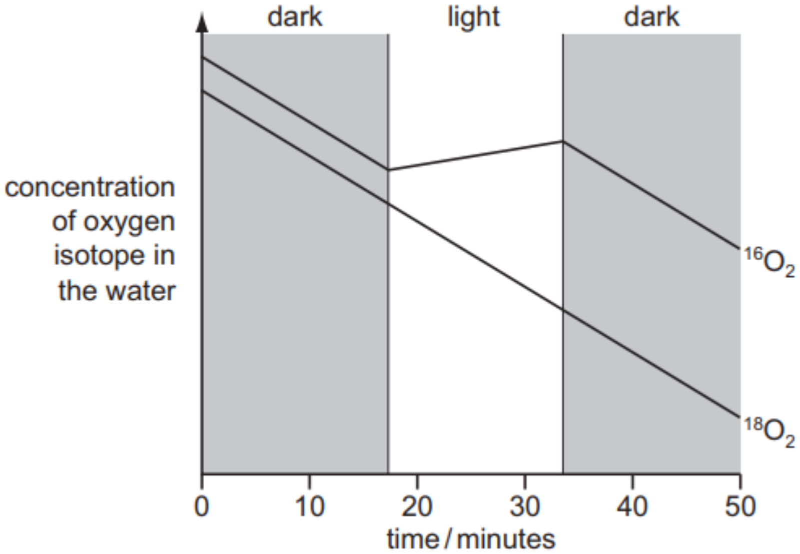
20 An aqueous suspension of isolated chloroplasts will produce oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which colours of light are required for maximum oxygen production?

	type of compound	colours of light
A	electron acceptor	blue and green
B	electron acceptor	blue and red
C	electron donor	blue and green
D	electron donor	blue and red

[Turn over

21 The common isotope of oxygen is ^{16}O. Air containing $^{16}O_2$ and $^{18}O_2$ was bubbled through a suspension of algae for a limited period. After this, the concentration of these two isotopes of oxygen in the water was monitored for the next 50 minutes whilst the algae were subjected to periods of dark and light. The results are shown in the diagram.



What is the best explanation for these results?

A Both isotopes of oxygen are used by the algae in the dark in respiration, but in the light oxygen is produced from water in photorespiration.

B The algae can distinguish chemically between the two isotopes.

C The algae produce oxygen from the water during photosynthesis, but only in the light.

D The two isotopes have different rates of diffusion.

22 Different types of reactions occur in the Calvin cycle.

Which statement correctly describes Calvin cycle?

A Carboxylation occurs in the conversion of triose phosphate to ribulose bisphosphate.

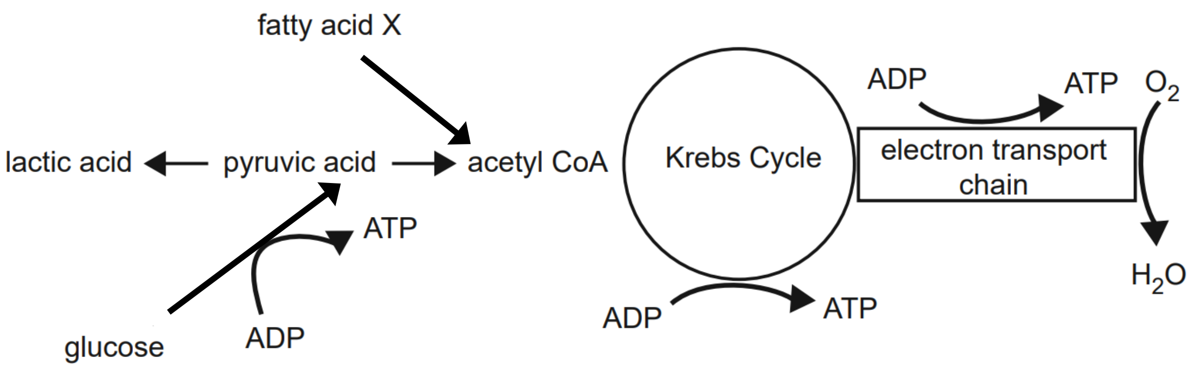
B Decarboxylation occurs in the conversion of ribulose bisphosphate to 3-phosphoglycerate.

C Phosphorylation occurs in the conversion of ribulose bisphosphate to 3-phosphoglycerate.

D Reduction occurs in the conversion of 3-phosphoglycerate to glyceraldehyde-3-phosphate.

23 If there is insufficient glucose for cellular respiration, fatty acids can be converted to acetyl CoA.

The diagram summarises the pathways for the breakdown of fatty acid X and glucose molecules. Each molecule of fatty acid X produces eight molecules of acetyl CoA.



Which statement is correct?

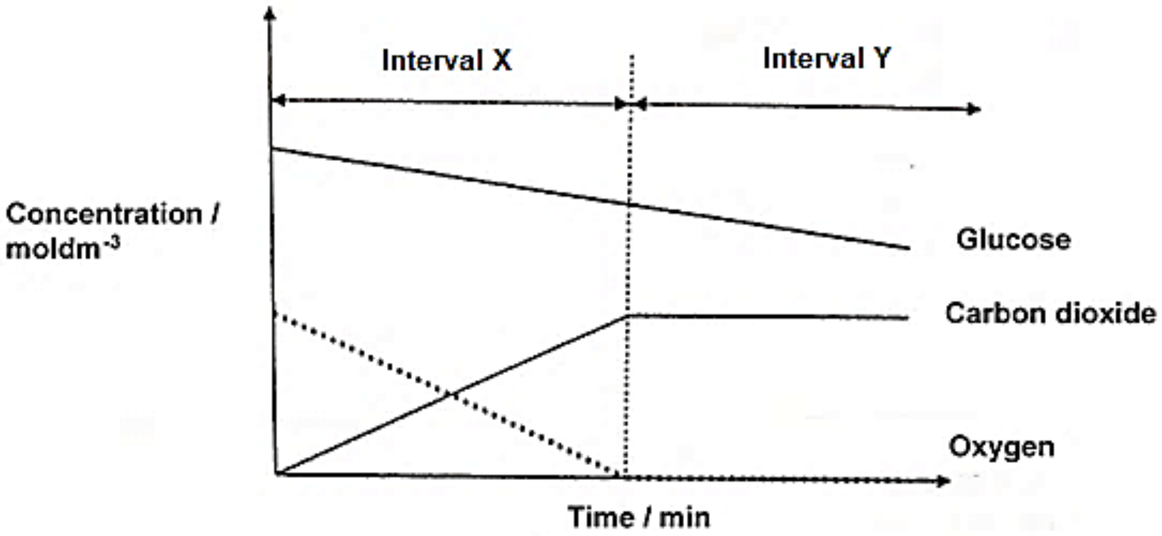
A For every glucose molecule oxidised, more ATP is made in the Krebs cycle than in glycolysis.

B Under aerobic conditions, one molecule of fatty acid X produces more ATP than one glucose molecule.

C Under anaerobic conditions, fatty acids are preferred over glucose as respiratory substrates.

D Oxygen plays a more important role as the final electron and proton acceptor when glucose molecules are oxidised as compared to the oxidation of fatty acids.

24 In an experiment, metabolically active cells were introduced to a sealed container of nutrient solution. The graph of the concentrations of glucose, carbon dioxide and oxygen were then analysed over time.



Which row identifies the processes happening at intervals X and Y?

	Interval X	Interval Y
A	Aerobic respiration	Lactic acid fermentation
B	Aerobic respiration	Alcoholic fermentation
C	Lactic acid fermentation	Aerobic respiration
D	Alcoholic fermentation	Aerobic respiration

25 The processes following an increase in blood glucose level are listed. The processes are **not** listed in the correct sequence.

- Activated protein kinase A phosphorylates and activates glycogen phosphorylase, which catalyses the breakdown of glycogen to glucose.
- Activated receptor binds to and activates a specific G protein located on the cytoplasmic side of the plasma membrane.
- Activated G protein then activates a membrane-bound enzyme.
- Activated adenyl cyclase converts ATP to cAMP, leading to an increase in concentration of cAMP.
- Binding of glucagon to a G-protein linked receptor causes the receptor to change 3D conformation.
- GTP nucleotide replaces the GDP bound to the G protein.
- cAMP acts as a second messenger and binds to and activates protein kinase A.

What is the consequence when the fourth step in the correct sequence of steps does not occur?

A ATP will be hydrolysed to ADP.

B Adenyl cyclase will be hyperactive.

C Protein kinase A will remain inactive.

D G protein will remain bound with GDP.

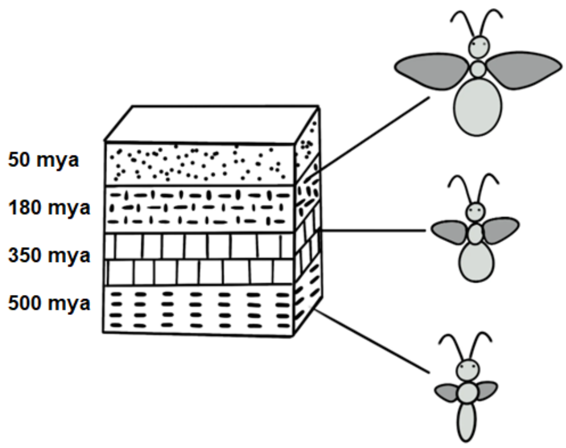
26 Several closely related frog species of the genus *Rana* are found in the forests of the South-eastern United States. The species boundaries are maintained by reproductive barriers. The statements below describe the different type of reproductive barriers that exist between the different species.

1. Males of one species sing only when its predators are absent; males of another species sing only when its predators are present.
2. One species mate at the season when daylight is increasing from 13 hours to 13 hours and 15 minutes; another species mate at the season when daylight is increasing from 14 hours to 14 hours and 15 minutes.
3. Two species of frogs belonging to the same genus occasionally mate, but the offspring fail to develop and hatch.

Which row correctly matches the statement to the type of reproductive barrier that is described?

	1	2	3
A	temporal	behavioural	hybrid inviability
B	seasonal	geographical	hybrid sterility
C	sympatric	temporal	gamete incompatibility
D	behavioural	temporal	hybrid inviability

27 A team of palaeontologists discovered the remains of an insect-like organism trapped in amber, which was found in rock layers believed to be about 180 million years old. As they dug deeper into the older layers of rock, they made several further discoveries. The diagram shows their findings.



Key:

mya: million years ago

Which statement can be concluded from the information provided?

A Speciation occurred gradually, with many small changes accumulating over a long period of time.

B The changes observed in these populations occurred gradually from 500 million years ago until 180 million years ago.

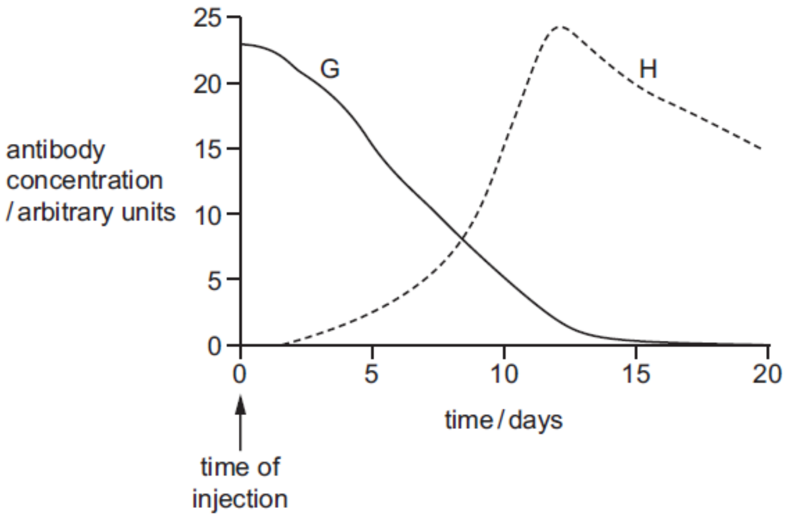
C The original population changed dramatically over time which lead to several different species forming.

D There were several large speciation events that occurred within a short period of time within these populations.

28 Tetanus is a bacterial infection.

The graph shows the blood antibody concentration of two people.

On day 0, person G was injected with antibodies to the tetanus toxin and person H was injected with the vaccine for tetanus.



What could be the result if G and H were infected with the tetanus bacteria on day 20?

A A second antibody peak would occur in person H that would be lower than the first peak.

B Tetanus antibodies would not be produced in person G.

C Antibody concentration would stay constant in person H.

D Antibody production would peak after day 32 in person G.

29 Some animals have genes that code for small peptides called cathelicidins. These peptides kill a wide range of bacteria by attaching to lipids in the bacterial membranes to weaken or disrupt them.

Scientists have produced a synthetic version of cathelicidin that kills bacteria which are resistant to a number of antibiotics such as tetracycline.

Which pair of statements explain how this synthetic cathelicidin might counter the problem of antibiotic resistance?

1. It is synthetic so bacteria can never become resistant to it.
2. It could be used instead of tetracycline, allowing tetracycline resistance to be reduced.
3. The only way a bacterium could develop resistance to it is by altering all the lipids in its membrane.
4. It could be used to kill multidrug-resistant strains of bacteria for which there is no effective antibiotic.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

[Turn over

30 The speckled wood butterfly (*Pararge aegeria*) is commonly found in woodland in southern parts of Britain. In the last 40 years, *P. aegeria* has significantly increased both its abundance and its ecological range in Britain. This is thought to be due to climate change, allowing the species to survive in more northerly habitats.

What impact would the increased ecological range of *P. aegeria* have on other species?

1 There could be increased food sources for birds.
2 More plants could be damaged by caterpillars.
3 There could be more pollination of flowers.
4 There could be increased competition with butterflies of other species.

A 1, 2, 3 and 4 **B** 1 and 4 only **C** 2 and 3 only **D** 2 and 4 only

Answers

1	A	**16**	D
2	D	**17**	D
3	C	**18**	A
4	D	**19**	D
5	C	**20**	B
6	C	**21**	C
7	A	**22**	D
8	B	**23**	B
9	D	**24**	A
10	B	**25**	C
11	B	**26**	D
12	B	**27**	B
13	C	**28**	D
14	A	**29**	D
15	D	**30**	A